Finite Differences: Summary

Derivatives from function values

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October 21, 2026

The task

Given only values of f

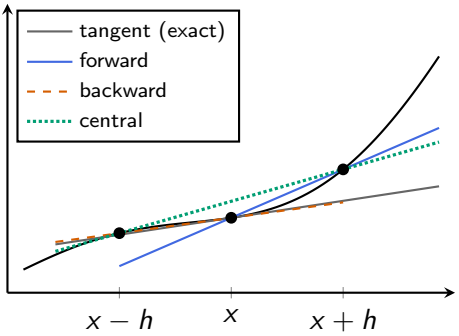
Approximate the slope at x:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- Take a **small step** h: a **secant** instead of the tangent.
- No formula for f' needed: only **evaluations** of f.

All formulas come from Taylor's theorem:

$$f(x \pm h) = f \pm hf' + \tfrac{h^2}{2} f'' \pm \tfrac{h^3}{6} f''' + \dots$$



The central secant is almost **parallel** to the tangent.

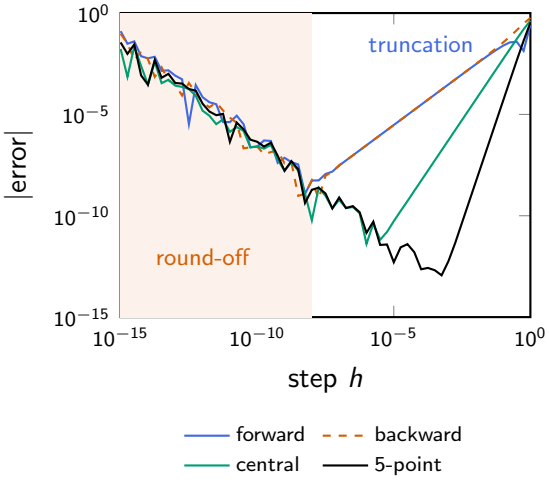
Four formulas

name	formula	error
forward	$\dfrac{f(x+h)-f(x)}{h}$	$O(h)$
backward	$\dfrac{f(x)-f(x-h)}{h}$	$O(h)$
central	$\dfrac{f(x+h)-f(x-h)}{2h}$	$O(h^2)$
5-point	$\dfrac{-f(x+2h)+8f(x+h)-8f(x-h)+f(x-2h)}{12h}$	$O(h^4)$

- forward / backward: error $\approx \pm\frac{h}{2}f''$, **opposite signs**
- central = their average $\Rightarrow$ the f'' terms **cancel**, error $\approx \frac{h^2}{6}f'''$
- 5-point: combine steps h and $2h$ $\Rightarrow$ the h^2 terms cancel too

Order $O(h^p)$: halve h $\Rightarrow$ error $\div 2^p$

Choosing h: two errors pull in opposite directions



Large h: truncation error

Error $\propto h^p$: falls as h shrinks.

Small h: round-off error

$f(x+h) \approx f(x)$: the subtraction loses digits.
Error $\propto 1/h$: grows as h shrinks.

Each curve is a **V**: the **best** h is at the bottom.

formula	best h	best error
forward, backward	10^{-8}	10^{-8}
central	10^{-5}	10^{-11}
5-point	10^{-3}	10^{-13}

Only a table of values: no function to call

Data on a grid, spacing $h = x_{i+1} - x_i$

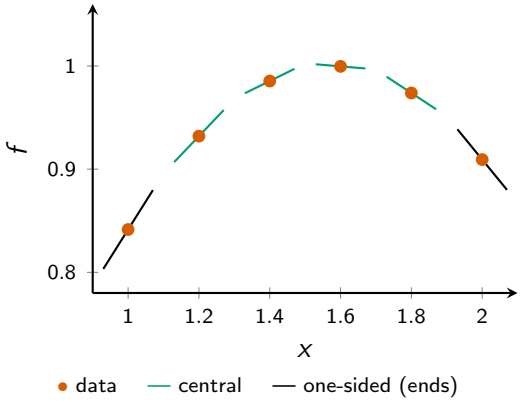
(x_0, f_0), (x_1, f_1), $\ldots$, (x_N, f_N): measurements or a file. We cannot evaluate $f(x + h)$ for a new h: h **is fixed by the data**.

Interior points $\rightarrow$ **central**, $O(h^2)$:

$$f_i' \approx \frac{f_{i+1} - f_{i-1}}{2h}$$

End points: no neighbour on one side $\rightarrow$ **one-sided 3-point**, also $O(h^2)$:

$$f_0' \approx \frac{-3f_0 + 4f_1 - f_2}{2h}, \qquad f_N' \approx \frac{3f_N - 4f_{N-1} + f_{N-2}}{2h}$$



Two ways to get a derivative

	finite differences	**symbolic** (SymPy)
what you need	values of f	a formula for f
step h?	yes	no
result	a number, with an error	a formula, exact
accuracy	10^{-8} to 10^{-13}	exact
works on a table of data?	**yes**	no
best for	tables, measured data	clean formulas

- Only a **table of values**? Finite differences are the tool: central inside, one-sided 3-point at the ends.
- Have a **formula**? Differentiate it symbolically, and check with a central difference.

Another option: **automatic differentiation** applies the derivative rules to the *code* of f.

Take-home

Finite differences

- Need only **values** of f, no formula.
- Forward, backward: $O(h)$. Central: $O(h^2)$. 5-point: $O(h^4)$.
- Symmetric formulas cancel error terms: use **central** or **5-point**.
- Only data? Central inside, one-sided 3-point at the ends.

Choosing h

- Too large: **truncation** error.
- Too small: **round-off** error.
- Best h is in between; never exact.
- Check against an exact derivative (SymPy) when you can.

A derivative from values: pick the formula, then pick h with care.